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MA3521 — In-Class Worksheet Monday/Tuesday Name:

$\mathbf{y}' = \boldsymbol{A}\mathbf{y} \quad \boldsymbol{A} = \begin{pmatrix} 2 & 1 & 1 \\ 0 & 2 & 0 \\ 0 & 0 & -1 \end{pmatrix} \quad \mathbf{y}(0) = \mathbf{y}_0 = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}$

Eigenvalues $\lambda_1 = \lambda_2 = 2$, $\lambda_3 = -1$ with $\mathbf{v}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ and $\mathbf{v}_3 = \begin{pmatrix} -1 \\ 0 \\ 3 \end{pmatrix}$ Note, only one eigenvector for $\lambda_1 = 2$.

1. The form for $\mathbf{y}_c$ is (use $\mathbf{w}$ for the additional vector)

$\mathbf{y}_c(t) =$

2. To find $\mathbf{w}$ you would row reduce

$\boldsymbol{M}_w =$

3. To satisfy the IC you would row reduce

$\boldsymbol{M}_{\text{IC}} =$

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Solve using variation of parameters

$$\mathbf{y}' = \boldsymbol{A}\mathbf{y} + \mathbf{f}(t) \qquad \boldsymbol{A} = \begin{pmatrix} -5 & 3 \\ -8 & 5 \end{pmatrix} \qquad \mathbf{f}(t) = \begin{pmatrix} 0 \\ 3e^t \end{pmatrix}$$

Eigenpairs 1, $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$ and -1, $\begin{pmatrix} 3 \\ 4 \end{pmatrix}$

1. $\mathbf{y}_c(t) =$ $\Phi(t) =$

$2\det \Phi(t) =$ $\Phi(t)^{-1} =$

3. $\mathbf{u}'(t) =$